

MOVIE RATING PREDICTION

Course replication and a timestamp extension

MovieLens 1M / October 7, 2026

Jiaming Cui, Joonse Lim, Tung Jerateepkulmeth

MOVIELENS AND RATING ACTIVITY

1,000,209

ratings from 6,040 users

95.74%

of user-catalog cells unobserved

Gaps between adjacent ratings within a user

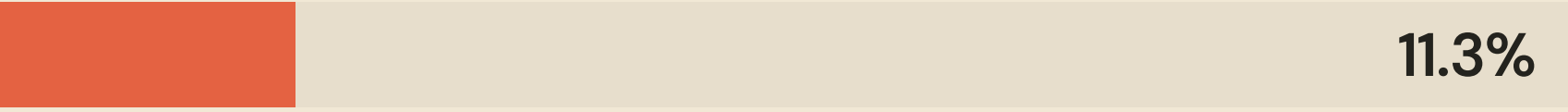
Same timestamp



1 to 60 seconds



More than 60 seconds



Rating order cannot establish viewing order.

Source: MovieLens 1M. All 994,169 within-user adjacent gaps.

COURSE BASELINE AND FRESH REPLICATION

CLASSROOM BENCHMARK

Exact 99/1 split, seed 144
20-fold CV selects rank 8

FRESH REPLICATION

400 R fits: 20 ranks × 20 folds
Fresh CV selects rank 9

Rank-8 test	RMSE	MAE	OSR ²
Instructor notebook	0.919	0.709	0.332
Our classroom refit	0.9198	0.7106	0.3300

Rank 8 retained for the classroom comparison. Original R seed/version are unrecorded.

RANDOM COMPLETION AND FUTURE PREDICTION

Same 1–5 rating target. The split follows the prediction question.

Partition	Random 99/1	Global time 80/10/10
Train	990,206	800,164
Validation	20–fold CV in train	100,024
Test	10,003	100,021

Random: complete missing ratings.
Global time: predict future ratings.

Keep timestamp ties together.
Refit train + validation before test.

METADATA AND TIME IMPROVE RATING ACCURACY

Ridge combines CF predictions and features. Lower RMSE is better.

Model	Random RMSE	Future RMSE
Training mean	1.1237	1.0893
Collaborative filtering	0.9198	1.2246
CF + metadata	0.8990	1.0303
CF + metadata + timestamp	0.8940	1.0001

Timestamp increment vs. metadata: 0.0050 random / 0.0302 future

TIMESTAMP FEATURES AND EVALUATION LIMITS

UTC calendar, elapsed days, and counts/gaps from earlier ratings.
Frozen fitted history. Equal-second entries excluded.

Timestamp vs. metadata	RMSE reduction	Paired 95% interval
Random	0.0050	[0.0031, 0.0069]
Future	0.0302	[0.0111, 0.0504]

Warm ratings: 23.84% in validation / 95.70% in final test.
This cohort change limits transfer of selected parameters.

Intervals condition on this split and selection. Future CF is worse than training mean.

FURTHER CHECKS BEFORE A BUSINESS PILOT

MOVIE STARTS

Proposed pilot measure:
recommendation-led starts
per exposed user.

Repeat evaluation across time windows before a limited randomized pilot.
MovieLens has no exposure, revenue or cost data to establish ROI.

Before a pilot

Evaluate Top-N ranking and cold users.

Candidate system

CF + metadata + timestamp,
with cold-user fallback.

Guardrails

Completion, catalog concentration,
latency and privacy.